Creating Strings

DESCRIPTION

You are given a word made of lowercase letters. By putting its letters in a different order you can make other words. Each letter must be used exactly as many times as it appears in the given word, and the given word itself also counts.

A letter may appear more than once in the word, so two different ways of ordering the letters can produce the same result. Such a result must be listed only once.

Find all the different results and list them in alphabetical order.

INPUT FORMAT

The input contains one line with a string s of length n. The string consists of lowercase letters `a` to `z` only.

OUTPUT FORMAT

On the first line, print an integer k: the number of different strings that can be made.

Then print the k strings, one per line, in alphabetical order (the same order that `strcmp` uses). End every line with a newline.

CONSTRAINTS

$1 \le n \le 8$

SAMPLE

Sample 1 Input

```
aabb
```

Sample 1 Output

```
6
aabb
abab
abba
baab
baba
bbaa
```

SOURCE

Adapted from the CSES Problem Set, task 1622.